# Backyard Birds & Disease: Pathogen Transmission at the Human Wildlife Interface

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## Background

- > 70% of diseases are zoonotic - originating in wildlife and capable of spreading to other species[1]
- Bird feeding & poultry keeping are widespread in the US[2]
- Bird feeders increase contact and fecal contamination[3]
- Backyard poultry can also be reservoirs for bacteria like *Salmonella enterica*[4]
- **We do not know the combined effects of feeders & poultry on disease in wild birds**

## Objectives

This study investigates how feeders and backyard poultry influence disease in wild birds.

- **Aim 1:** Test relationship between food visitation frequency and pathogen prevalence using RFID
- **Aim 2**: Compare pathogen prevalence among sites with bird feeders and/or poultry

## Experimental Design

12 study sites sampled monthly in central Alabama (June 2024–2027)

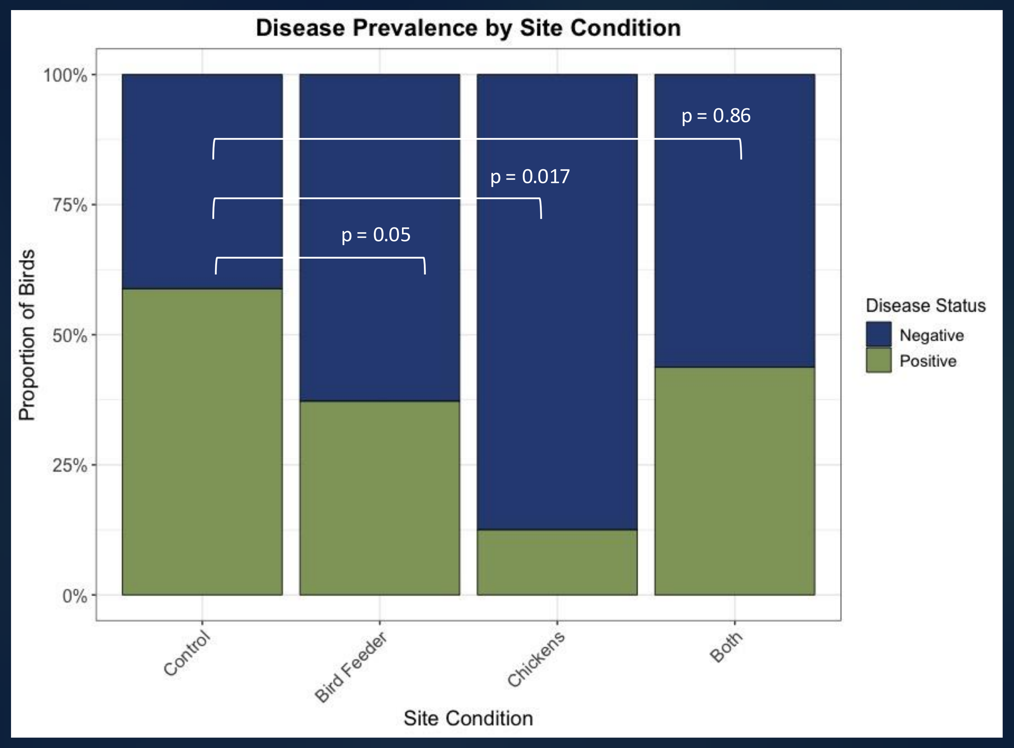
Four treatment types:
- Control sites (no feeders/poultry): n = 4
- Bird feeders only: n = 4
- Backyard chickens only: n = 2
- Both feeders and chickens: n = 2

**Field Sampling**
- Wild songbirds captured using mist nets
- >230 birds captured to date
- Common resident species receive PIT tags to track food visitation via RFID
- Collected wild bird & chicken fecal samples

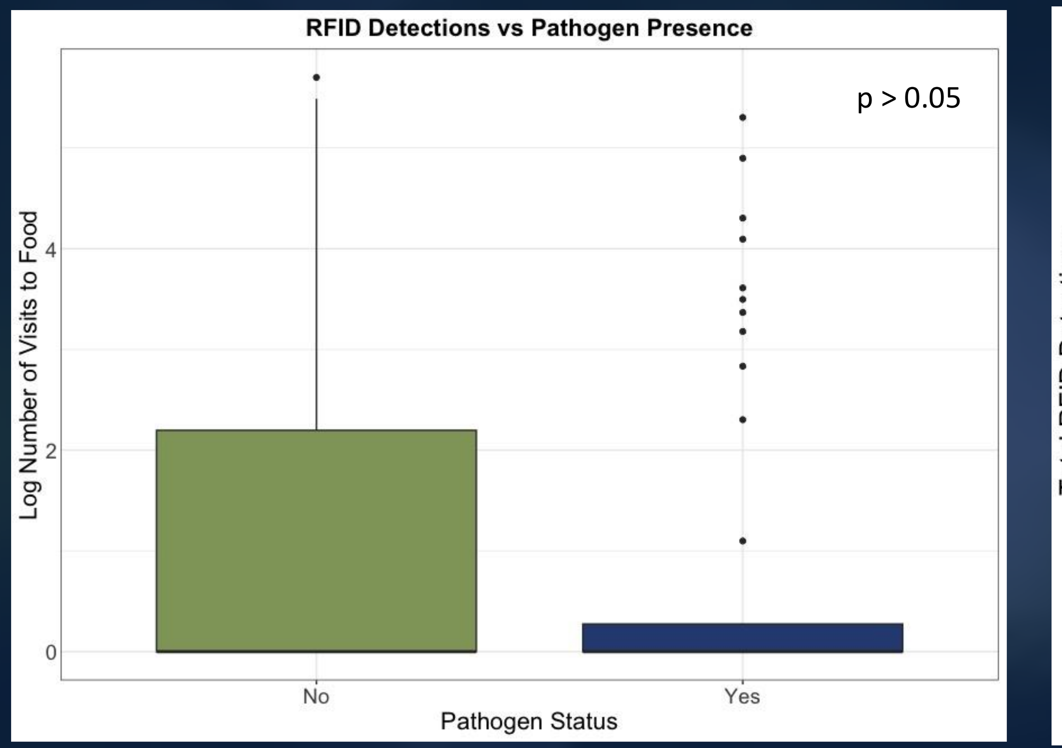
**Laboratory Analysis -** Samples cultured for bacterial pathogens

## Methods Figures

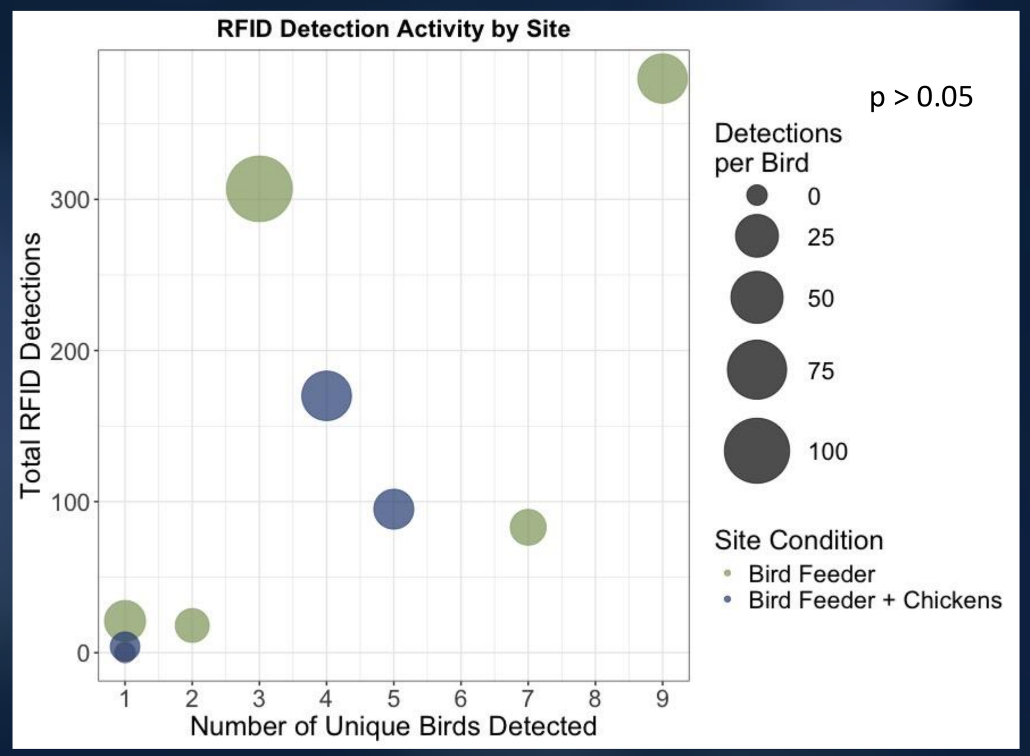

## Results




More birds had detectable bacterial infections at Control sites.




RFID detection rates did not differ significantly between infected and uninfected birds.




Bird feeder and bird feeder + chicken sites showed similar RFID detection patterns.

## Conclusions

- Supplemental food was correlated with reduced bacterial infections in wild birds
- Additional food may improve the bird's immune system[5]
- In the future we hope to:
  - Expand sampling at chicken-keeping households to increase statistical power
  - Improve RFID antennas to increase detections
  - Develop evidence-based biosecurity guidelines for backyard practices

## Acknowledgements

We thank all the volunteers for their hard work and homeowners for permitting us to work in their backyards!

## References and Contact Information

1. Jones et al., 2008, Nature; 2. U.S. Fish & Wildlife Service, 2022, Natl. Surv. Fish. Hunt. Wildl.-Assoc. Rec.; 3. Becker et al., 2015, Ecol. Lett.; 4. Larsen et al., 2022, Front. Vet. Sci.; 5. Owen et al., 2021, Proc. R. Soc. B

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